

Zhiyao He

Email: zhiyaohe@outlook.com | Webpage: zhiyaohe.github.io | GitHub: github.com/zhiyaohe

Robot Learning · Reinforcement Learning · Vision-Language-Action Models · SLAM · Robotics Simulation

Profile

I am an AI engineer working on embodied AI systems. I build robot pipelines that connect perception, language models, planning and control, from simulation (Gazebo, MuJoCo, Isaac Gym) to real hardware. My recent work is a voice-controlled navigation and manipulation system on the Unitree Go2-W quadruped robot. My goal is to do a PhD in robot learning, especially how robots can learn to perceive, reason, and act reliably in real-world environments.

Education

M.S. in Signal and Information Processing Hohai University, Nanjing, China Sep 2015 to Jul 2018

Honors: Outstanding Graduate Student (Research-Based Degree)

- Thesis work on machine learning for runoff forecasting; published two peer-reviewed journal papers and obtained one Chinese invention patent.
- Relevant courses: Information Acquisition and Processing, Optimization. Certificate: Deep Learning Specialization (DeepLearning.AI).

B.Eng. in Electrical Engineering and Automation Anhui Agricultural University, Hefei, China Sep 2011 to Jul 2014

CGPA: 3.65 / 4.00

- Finished the four-year program in three years - the only student in my cohort to do so.
- Relevant courses: Control Theory, Circuit Design, C Programming, Digital Signal Processing.

A.S. in Computer Science University of the People, USA Nov 2021 to Nov 2024

CGPA: 4.00 / 4.00

- President's List for 12 terms.
- Relevant courses: Python, Java, Databases, Operating Systems, Software Engineering, Computer Networks.

Standardized Tests

TOEFL: 5 / 6 (equivalent to 100 / 120), CEFR C1, Sep 2026

Publications and Patent

- **Zhiyao He**, Mingxiang Yang, Hao Wang, et al. "Annual Average Runoff Ensemble Forecast for Jinping I-Stage Hydropower Station Based on Elman Neural Network." Water Resources and Power, 2017.
- Chenming Li, Lei Zhu, **Zhiyao He**, et al. "Runoff Prediction Method Based on Adaptive Elman Neural Network." Water, 2019.
- **Zhiyao He** (Patent holder). "Medium- and Long-term Runoff Forecasting Method Based on Mutual Information, Kernel PCA, and Elman Neural Network." Chinese National Invention Patent, CN107463993B, granted 2020.

Work Experience

AI Engineer WellWin Technology Co., Ltd., Hengqin & Macau, China Mar 2026 to Present

- Built the perception and grasping pipelines for voice-controlled navigation and manipulation on the Unitree Go2-W quadruped robot with a D1 arm. This was a team project (details in Projects).
- Built a fully offline agent for the AgiBot Lingxi X2 robot on Jetson (ASR, LLM, RAG, TTS) that stays reliable during live shows and other events where network signal is unstable.
- Built an online agent for the AgiBot Lingxi X2 robot and integrated it with third-party business systems, so the robot can pull and act on live backend data.
- Set up simulation environments and ran locomotion training for the AKI and Unitree G1/R1 humanoid robots in Gazebo, MuJoCo and Isaac Gym.
- AKI humanoid robot (a company education product; team reproduction of Unitree's open-source Qmini): responsible for the control system, power system, code debugging, and simulation training.
- Designed and built a full humanoid robotics curriculum for Macau secondary schools on the AgiBot Lingxi X2, covering three platforms (LingXin conversational agent, LinkCraft no-code choreography, AimDK ROS 2 SDK) and four grade-level extension modules of rising difficulty, from "X2 Says Hello" to "X2 as a Smart Service Robot".

Independent Study and Research Nanjing, China Jul 2019 to Feb 2026

- After Huawei, I planned to return to research. During this period my second journal paper (Water, 2019) was published and my invention patent was granted (2020).
- From 2021 to 2024, I completed a full computer science degree at University of the People (GPA 4.00/4.00, see Education) to strengthen my foundation in computer science and dive deeper into AI. Part of this period overlapped with the COVID-19 lockdowns in China.
- Admitted to the M.Sc. in Data Science at Trinity College Dublin (2025 intake), but declined the offer for funding reasons.
- After declining the offer, I prepared for AI engineering interviews and joined WellWin Technology in March 2026.

Software Engineer Huawei Technologies Co., Ltd., Nanjing R&D Institute, China Jul 2018 to Jun 2019

- Worked in the intelligent security systems team on face, vehicle, and other recognition products. My work covered system development, deployment, and feature validation (not algorithm development).
- Cross-team debugging and system-level testing.

Research Intern China Institute of Water Resources and Hydropower Research, Beijing, China Jun 2016 to Jun 2017

- Built a complete runoff prediction pipeline: data collection and cleaning, feature engineering, model development, hyperparameter tuning, training and evaluation.
- This work produced two journal papers and one invention patent (listed above).

Selected Projects

Voice-Controlled Navigation and Manipulation on Unitree Go2-W team project with Macau Millennium College

- Perception (my work): fine-tuned YOLO11 by transfer learning to detect a specific bottle, using a wrist-mounted depth camera and 2D-to-3D coordinate transformation to locate the target in the world frame.
- Manipulation (my work): a grasping pipeline built on MoveIt 2 (KDL inverse kinematics, OMPL/RRTConnect planning, FCL collision checking) that plans a collision-free path and executes the grasp.
- Navigation and language (teammates): SLAM-based navigation and the ASR/LLM/OpenClaw/TTS pipeline; we defined the interfaces and integrated the full system together.
- Developed in ROS 2, tested in MuJoCo and Gazebo simulation and on the real robot.

AKI Humanoid Robot company education product; team reproduction of Unitree's open-source Qmini

- My parts: brought up and debugged the control system and power system on the reproduced robot, and ran the RL locomotion training in Isaac Gym. Hardware and assembly were shared team work.
- ASR/LLM/TTS pipeline for natural voice interaction.

Voice RAG Assistant on AgiBot Lingxi X2 industry project for CTM (Macau Telecom)

- An in-store retail assistant: customers ask about products, plans, and pricing by voice.
- Built the full pipeline to run offline on a Jetson: a bridge node streams the X2's microphone audio to a local FastAPI backend, which runs local ASR (FunASR/Paraformer-zh), a local LLM (Qwen2.5-7B via Ollama), and pgvector RAG over the product database (PostgreSQL); answers are spoken through the X2's built-in offline TTS.

Humanoid Robotics Curriculum on AgiBot Lingxi X2 education product for Macau junior and senior secondary schools

- Architected a 20-lesson core course and four grade-level modules as a spiral curriculum that progresses from no-code tools (LingXin, LinkCraft) to ROS 2 programming with the AimDK SDK, and finally to multi-platform integration.
- Designed and built the four grade-level modules, each culminating in a ROS 2 capstone that runs on the real robot, as themed scenarios of rising difficulty: "X2 Says Hello" (preset arm motions), "X2 Expresses" (screen, LED, speech), "X2 in Motion" (bipedal locomotion), and "X2 as a Smart Service Robot" (sensing and full-chain integration).
- Engineered the four capstones on the real robot; the senior-secondary one links a head touch to a facial emotion, a spoken greeting, an arm wave, and a walk. Making it reliable meant resolving a single-threaded-executor deadlock, service-priority conflicts that dropped the speech and expression commands, and the control handoff between the robot's walking mode and the stance its arm gestures require.

Technical Skills

- Programming & Systems: Python, C++, Linux, Docker, Git
- Robotics: ROS/ROS 2, MoveIt 2, Nav2, Gazebo, MuJoCo, Isaac Gym, SLAM, Navigation, Perception
- AI & Robot Learning: PyTorch, Machine Learning, Computer Vision, Reinforcement Learning, LLMs, RAG, VLA
- Robot Platforms: Unitree G1, Unitree B2/Go2/Go2-W, AgiBot Lingxi X2

Research Interests

- Robot Learning and Reinforcement Learning
- Vision-Language-Action Models and World Action Models
- Robot Manipulation and Grasping
- Robotics Simulation, SLAM and Navigation

Honors and Awards

- Outstanding Graduate Student, Hohai University | 2017
- Hohai University Scholarship | 2015 to 2017
- Anhui Agricultural University Scholarship | 2012
- Outstanding Contribution Award, Huawei Nanjing R&D Institute | 2019
- President's List, University of the People (12 terms) | 2021 to 2024